

AMTCH - Parag Sir

* SET: $A = \{1, 2\}$; $B = \{1, 2, 2\}$; $C = \{1, 2, 3\}$
 $\downarrow$
 $(G, \oplus)$ A and B are same sets.

* Binary Operator: - If G is a non-empty set and $\otimes$ is defined on G then $\otimes$ is said to be a binary operator iff:-

$\otimes: G \times G \rightarrow G$: This means that cross maps $G \times G$ to G

Example: $G = \mathbb{R}$ & $\otimes = +$. Then $+$ is a binary operator.

PROPERTIES:-

- Commutative: $a \otimes b = b \otimes a \quad \forall a, b \in G$

Eg. $G = \mathbb{R}$; $\otimes = +$ ✓

$G = \mathbb{R}$; $\otimes = \times$ ✓

$G =$ set of matrices ; $\otimes = \times$ Not commutative

- Associative: $a \otimes (b \otimes c) = (a \otimes b) \otimes c$

- Distributivity: $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$

↳ Left distributivity.

$(a \oplus b) \otimes c = (a \otimes c) \oplus (b \otimes c)$ → Right Distributivity

- Identity Element: - An element $e \in G$ is said to be left identity element iff: $e \otimes a = a \quad \forall e, a \in G$

- Inverse of an element: - b will be the inverse of a iff:

$b \otimes a = e$ → Left inverse

$a \otimes b = e$ → Right inverse

* GROUP :- A group is an algebraic structure $(G, \otimes)$ which satisfies the following:-

- ① $a \otimes (b \otimes c) = (a \otimes b) \otimes c \quad \forall a, b, c \in G$
- ② $\forall a \in G \exists e \text{ s.t. } a \otimes e = e \otimes a = a$
- ③ $\forall a \in G \exists b \text{ s.t. } a \otimes b = b \otimes a = e$

Q) If G is a group s.t. $(a \otimes b)^2 = a^2 b^2 \quad \forall a, b \in G$ then prove that G is **Abelian group** (should be commutative)

$$\begin{aligned} (a \otimes b)^2 = a^2 b^2 &\Rightarrow (a \otimes b) \otimes (a \otimes b) = (a \otimes a) \otimes (b \otimes b) \\ \text{Let } c &= a \otimes b \in G; \text{ Then L.H.S} \rightarrow c \otimes (a \otimes b) \\ &= (c \otimes a) \otimes b = [(a \otimes b) \otimes a] \otimes b \\ &= [a \otimes (b \otimes a)] \otimes b \end{aligned}$$

$$\begin{aligned} \text{Assuming } b \otimes a &= a \otimes b \rightarrow (a \otimes (a \otimes b)) \otimes b \\ &= ((a \otimes a) \otimes b) \otimes b = (a \otimes b) \otimes b = a \otimes (b \otimes b) \\ &= (a \otimes a) \otimes (b \otimes b) = \text{RHS} \end{aligned}$$

Hence our assumption is correct. G is an Abelian group.

* GROUPS, RINGS & FIELDS

Q) Verify whether a set of $n \times n$ non singular matrices with real elements forms a group under matrix addition.

- Let $A_{n \times n}$ be a non singular matrix with real elements.
 $(G, +)$

$$G = \left\{ \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & & a_{nn} \end{bmatrix} : a_{ij} \in \mathbb{R}, n \in \mathbb{N} \right\}$$

$$\det(G) \neq 0$$

- Test of Associativity : Consider $A, B, C \in G$

$$\begin{aligned} A + (B + C) &= \begin{bmatrix} a_{11} + (b_{11} + c_{11}) & \dots & a_{1n} + (b_{1n} + c_{1n}) \\ a_{21} + (b_{21} + c_{21}) & & a_{2n} + (b_{2n} + c_{2n}) \\ \vdots & \vdots & \vdots \end{bmatrix} \\ &= \begin{bmatrix} (a_{11} + b_{11}) + c_{11} & \dots & (a_{1n} + b_{1n}) + c_{1n} \\ (a_{21} + b_{21}) + c_{21} & & (a_{2n} + b_{2n}) + c_{2n} \\ \vdots & \vdots & \vdots \end{bmatrix} \\ &= (A + B) + C \quad \checkmark \end{aligned}$$

- Test of identity :- Consider $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ b_{21} & \dots & b_{2n} \\ \vdots & \vdots & \vdots \end{bmatrix} \in G$

Let $E = \begin{bmatrix} e_{11} & \dots & e_{1n} \\ e_{21} & \dots & e_{2n} \\ \vdots & \vdots & \vdots \end{bmatrix} \in G$. If possible $[a_{ij}, e_{ij} \in \mathbb{R}]$

$$\text{If } A + E = A \quad \text{then : } \begin{bmatrix} a_{11} + e_{11} & \dots & a_{1n} + e_{1n} \\ a_{21} + e_{21} & \dots & a_{2n} + e_{2n} \\ \vdots & \vdots & \vdots \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \vdots & \vdots \end{bmatrix}$$

For two matrices to be equal, their corresponding elements must be equal $\Rightarrow$ For ij^{th} element
 $a_{ij} + e_{ij} = a_{ij} \Rightarrow e_{ij} = 0$

$$\Rightarrow E = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & 0 \end{bmatrix}_{n \times n} \quad (\text{Null matrix})$$

But $\det(E) = 0 \Rightarrow E \notin G$
 This contradicts our assumption

Identity element does not exist

$\Rightarrow (G, +)$ is not a group.

Modification: Let G now be the set of all $n \times n$ matrices whether singular or not.

Then, identity element E exists and $E \in G$.

- Test for Inverse:

$$\text{Let } A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ b_{21} & \dots & b_{2n} \\ \vdots & & \vdots \end{bmatrix} \in G$$

s.t. $A + B = E$

$$E = \begin{bmatrix} 1 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{bmatrix}_{n \times n} \in G \rightarrow \text{already proven}$$

$$\therefore \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & \dots & a_{2n} + b_{2n} \\ \vdots & & \vdots \end{bmatrix} = \begin{bmatrix} 1 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{bmatrix}_{n \times n} \Rightarrow \text{For } ij^{\text{th}} \text{ element } a_{ij} + b_{ij} = 0$$

$$\Rightarrow b_{ij} = -a_{ij} \quad \forall i, j \in R$$

So, $B = \begin{bmatrix} -a_{11} & \dots & -a_{1n} \\ \vdots & & \vdots \\ -a_{n1} & \dots & -a_{nn} \end{bmatrix} \Rightarrow B \in G$ (as all its elements are real)
Hence **INVERSE EXISTS**.

Thus $(G, +)$ is a group when $G = \text{set of all } n \times n \text{ real matrices.}$

* RINGS:- A ring is an algebraic structure $(R, +, *)$ which satisfies the following:-

① $(R, +)$ is an **Abelian** group $\rightarrow$

- a) $(a+b)+c = a+(b+c) \quad \forall a, b, c \in R$
- b) $\forall a \in R \exists e \in R \text{ s.t. } a+e = e+a = a$
- c) $\forall a \in R \exists b \in R \text{ s.t. } a+b = b+a = e$
- d) $\forall a, b \in R, a+b = b+a$

② $(R, *)$ is a Monoid :-

- a) $(a*b)*c = a*(b*c) \quad \forall a, b, c \in R$
- b) Identity : $a*e' = e'*a = a \quad [\forall a \in R, \exists \text{ such } e' \in R]$

③ Distributivity of $*$ over $+$: $(a+b)*c = (a*c) + (b*c)$
 $a*(b+c) = (a*b) + (a*c)$

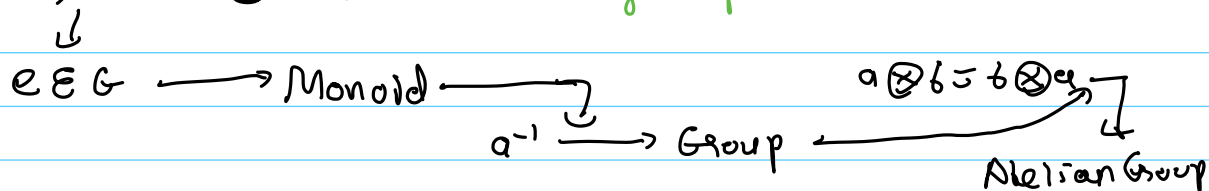
Not included in this definition of a ring

a) Multiplicative Inverse

b) Commutativity of Multiplication

→ If b is included, we get a field, but M.I. still unsure.

Note: ① $a \otimes (b \otimes c) = (a \otimes b) \otimes c \rightarrow$ Semi group



② Sub Group: $(G, \otimes)$
 $G \otimes G \rightarrow G$
 $H \otimes H \rightarrow H$

Simply taking a subset of the group's set does not give a subgroup. Identity and Inverse may not be in it.

Q> Verify whether a set of polynomials with real coeff in a ring?

$$(R, +, *) , R = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in R, n \in \mathbb{N} \cup \{0\} \right\}$$

① Checking if $(R, +)$ is Abelian $\rightarrow$ a) Let $A, B, C \in R$

$$\begin{array}{l} A = a_0 + a_1 x + \dots + a_n x^n \\ B = b_0 + b_1 x + \dots + b_n x^n \\ C = c_0 + c_1 x + \dots + c_n x^n \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} a_i, b_i, c_i \in R$$

$$\text{Now, } A + B = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n$$

$$\Rightarrow (A+B)+C = [(a_0+b_0)+c_0] + [(a_1+b_1)+c_1]x + \dots + [(a_n+b_n)+c_n]x^n$$

Since associativity for real no-s is known,

$$[(a_i + b_i) + c_i] x^i = [a_i + (b_i + c_i)] x^i$$

Hence Associativity is proven for $(R, +)$

Rest can be proven.

16/08/23

* "LINEAR" "VECTOR" "SPACES"

① $\Rightarrow$ Let V be a set of ordered pairs (a, b) , $a, b \in \mathbb{R}$
The following operations are defined

(i) $(a, b) \otimes (c, d) = (a+c, b+d)$

(ii) $k \otimes (a, b) = (ka, 0)$, $k \in \mathbb{R}$

Identify whether V is a vector space over $\mathbb{R}$

- An algebraic structure $(V, F, \otimes, \odot)$ equipped with a non-empty set V , field F , a binary operation $\otimes: V \times V \rightarrow V$, and an external mapping $\odot: F \times V \rightarrow V$ which satisfies:

- ① $(V, \otimes)$ is an Abelian group
- ② $a \odot (u \otimes v) = (a \odot u) \otimes (a \odot v)$ $\left. \begin{array}{l} u, v \in V \\ a, b \in F \end{array} \right\}$
- $(u \otimes v) \odot b = (u \odot b) \otimes (v \odot b)$

$\hookrightarrow$ Left Distributivity

$\hookrightarrow$ Right distributivity

$\odot$ is distributive over $\otimes$



Here, we consider F as an external set and not V , because:

- ① output is in V
- ② V is our set of interest

THIS ALTOGETHER MAKES A SPACE

- In a space, elements depend on other elements, contrary to a set, where elements are well defined.

(2)

$$V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \cdot x_i \in \mathbb{R}, n \in \mathbb{Z}^+ \right\}$$

$$\otimes = +$$

$$\odot = \times$$

$$F = \mathbb{R}$$

Is this a vector space?

ans) $u, v, w \in V$ s.t.

(i) $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad u_i, v_i, w_i \in \mathbb{R}$

$$(u+v)+w = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ \vdots \\ u_n+v_n \end{bmatrix} \overset{\text{Associative}}{\oplus} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} u_1+v_1+w_1 \\ \vdots \\ u_n+v_n+w_n \end{bmatrix}$$

$$u+(v+w) = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \oplus \begin{bmatrix} v_1+w_1 \\ \vdots \\ v_n+w_n \end{bmatrix} = \begin{bmatrix} u_1+v_1+w_1 \\ \vdots \\ u_n+v_n+w_n \end{bmatrix}$$

Hence

$$(u_i + v_i) + w_i = u_i + (v_i + w_i) \rightarrow \text{Associativity of Real Numbers.}$$

(ii)

$$u_i + e_i = u_i$$

$$\Rightarrow e_i = 0$$

$$\text{where } e_i \in E = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

$\rightarrow$ Identity element exists

(iii)

Inverse exists (try yourself)

Binary

$$(iv) \quad a \otimes (u_i + v_i) \quad a \in F \quad u_i, v_i \in V$$

$\downarrow \quad \downarrow \quad \downarrow$
 $\in R \quad \in R \quad \in R$

$$= (a \otimes u_i) + (a \otimes v_i) \rightarrow \text{Distributivity}$$

Similar proof for right distributivity.

- Now, if this exists, then the elements of set V is called a vector. $\rightarrow$ ordered n -tuple.
 $\swarrow$
dimensions

- Notation: vector V , Matrix M

- Hence if we say "vector spaces", we mean to say a set of vectors, which need not necessarily mean column matrices.

③

$$\begin{cases} f(x) = 2x \\ g(x) = \sin(x) \end{cases}$$

By plotting graphs, we can conclude linearity

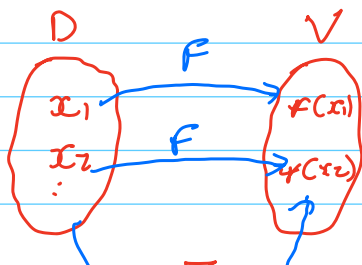
$$\rightarrow y = x^2 + e$$

$$dy/dx = 2x \rightarrow \text{Linear, if we don't solve}$$

$$dy/dx = y^2 \sin(x)$$

$$\begin{aligned} \text{Now, let } f(x_1) &= 2x_1, & f(x_2) &= 2x_2 \\ f(x_1 + x_2) &= 2x_1 + 2x_2 \\ &= f(x_1) + f(x_2) \end{aligned}$$

$$f(cx_1) = c f(x_1)$$



DEF^N OF
LINEARITY

Now

Let $\frac{dy}{dx} = 2y$

Linear without solving
Non linear after you solve $y = Ce^{2x}$

$$\frac{d}{dx} C(y_1 + y_2) = 2(y_1 + y_2) = \frac{dy_1}{dx} + \frac{dy_2}{dx}$$

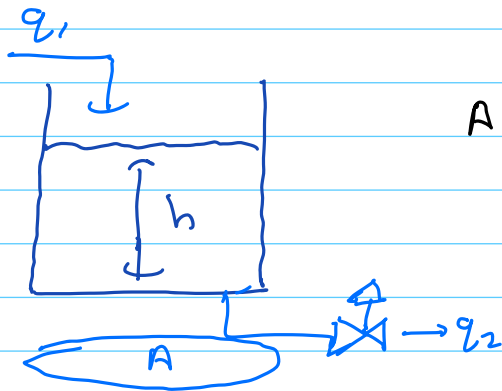
LINEAR

$$\frac{d}{dx} C(y_1) = 2cy_1 = C(2y_1) = C \frac{dy_1}{dx}$$

- Linearity is defined for OPERATORS and not for MEMBERS OF SET.

Note: Euclidean Space $\Rightarrow n=3$

eg)



$$A \left(\frac{dh}{dt} \right) = q_1 - q_2 \quad (\text{Mass Balance})$$

$$\frac{dh}{dt} = \frac{1}{A} f(h)$$

Levels of the liquid in the tank lie in a linear vector space.

Coming back to the original question

Let $v = (a_1, b_1)$, $w = (a_2, b_2)$, $u = (a_3, b_3) \in V$
 $a, b \in \mathbb{R}$

① To check if $(V, \otimes)$ is an Abelian group

(i) $(v \otimes w) \otimes u$

$$= \overset{\rightarrow C_1, C_2}{\left(\begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix} \right) \otimes w} = \begin{pmatrix} a_1 + a_2 + a_3 \\ b_1 + b_2 + b_3 \end{pmatrix} \quad \swarrow \text{Associativity}$$

$$v \otimes (w \otimes u) = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} \otimes \begin{pmatrix} a_2 + a_3 \\ b_2 + b_3 \end{pmatrix} = \begin{pmatrix} a_1 + a_2 + a_3 \\ b_1 + b_2 + b_3 \end{pmatrix}$$

(ii) Identity element

$$v \otimes e = v \quad \overset{\rightarrow e \in R}{e \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}} \rightarrow \text{Identity element exists}$$

(iii) Inverse element

$$v \otimes v = e \\ \begin{pmatrix} a_1 + a_2, b_1 + b_2 \end{pmatrix} = \begin{pmatrix} 0, 0 \end{pmatrix} \\ \Rightarrow v = \begin{pmatrix} -a_1, -b_1 \end{pmatrix} \rightarrow \text{Inverse exists}$$

② To check if distributivity holds

$$a \otimes (v \otimes w) = (a \otimes v) \otimes (a \otimes w)$$

$\hookrightarrow$ LHS

$$\Rightarrow k \otimes \begin{pmatrix} a_1 + a_2, b_1 + b_2 \end{pmatrix} = \begin{pmatrix} k(a_1 + a_2), 0 \end{pmatrix}$$

$$\begin{aligned} \text{RHS: } & [k \otimes \begin{pmatrix} a_1, b_1 \end{pmatrix}] \otimes [k \otimes \begin{pmatrix} a_2, b_2 \end{pmatrix}] \\ & = [k \otimes \begin{pmatrix} a_1, b_1 \end{pmatrix}] \otimes [k \otimes \begin{pmatrix} a_2, b_2 \end{pmatrix}] = \begin{pmatrix} k a_1, 0 \end{pmatrix} \otimes \begin{pmatrix} k a_2, 0 \end{pmatrix} \\ & = \begin{pmatrix} k(a_1 + a_2), 0 \end{pmatrix} \end{aligned}$$

HENCE LHS = RHS $\Rightarrow$ DISTRIBUTIVE

$\Rightarrow V$ is a vector space over R

- Let S be the set of all elements of the form

$$(x+2y, y, x+3y) \in \mathbb{R}^3 \quad x, y \in \mathbb{R}$$

Prove that S is a subspace of $\mathbb{R}^3$

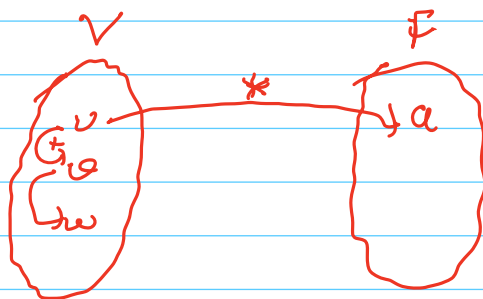
ans) Let $u = (x_1 + 2y_1, y_1, -x_1 + 3y_1) \quad x_1, y_1 \in \mathbb{R}$
 $v = (x_2 + 2y_2, y_2, -x_2 + 3y_2) \quad x_2, y_2 \in \mathbb{R}$

① $u + v \in S \Rightarrow (x_1 + x_2 + 2(y_1 + y_2), y_1 + y_2, -x_1 - x_2 + 3(y_1 + y_2))$
 $\downarrow \quad \downarrow$
 $x_3 \in \mathbb{R} \quad y_3 \in \mathbb{R} \Rightarrow u + v \in S$

② $cu \in S \Rightarrow (cx_1 + 2cy_1, cy_1, -cx_1 + 3cy_1)$ *Associativity & Commutativity of Real Nos.*
 $\downarrow \quad \downarrow$
 $x_4 \in \mathbb{R} \quad y_4 \in \mathbb{R} \Rightarrow cu \in S$

23/8/23

* LINEAR COMBINATION OPERATION



$$+ : V \times V \rightarrow V$$

$$* : F \times V \rightarrow V$$

$$w = au + bv$$

↳ Linear Combination

Q) Express the polynomial $f(x)$ as a L.C. of ϕ_i 's

$$f(x) = x^2 + 4x - 3$$

$$\phi_1(x) = x^2 - 2x + 5$$

$$\phi_2(x) = 2x^2 - 3x$$

$$\phi_3(x) = x + 3$$

$$\text{Let } f = a\phi_1 + b\phi_2 + c\phi_3 \quad (a, b, c \in F)$$

$$\text{So, } x^2 + 4x - 3 = a(x^2 - 2x + 5) + b(2x^2 - 3x) + c(x + 3)$$

Comparing coeff we get

$$1 = a + 2b \quad 4 = -2a - 3b + c \quad -3 = 5a + 3c$$

$$\begin{pmatrix} 1 & 2 & 0 \\ -2 & -3 & 1 \\ 5 & 0 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$$

$$\text{Solving } a = -3, b = 2, c = 4$$

Q) Consider $\underline{v}_1 = [1, 2, 3]^T$, $\underline{v}_2 = [2, 3, 1]^T$, $\underline{v} = [a, b, c]^T$. Find the conditions on a, b, c so that $\underline{v}$ is a L.C. of $\underline{v}_1$ and $\underline{v}_2$

$$\text{Let } \underline{v} = p\underline{v}_1 + q\underline{v}_2 \quad p, q \in F$$

$$\text{So } \begin{pmatrix} a \\ b \\ c \end{pmatrix} = p \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + q \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

$$\Rightarrow a = p + 2q \quad b = 2p + 3q \quad c = 3p + q$$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \text{Augmented: } \left(\begin{array}{cc|c} 1 & 2 & a \\ 2 & 3 & b \\ 3 & 1 & c \end{array} \right) \quad \begin{array}{l} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \end{array}$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b - 2a \\ 0 & -5 & c - 3a \end{array} \right) \quad R_3' = R_3 - 5R_2$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b-2a \\ 0 & 0 & 7a-5b+c \end{array} \right) \quad R_1' = R_1 - 2R_2$$

$$\left(\begin{array}{cc|c} 1 & 0 & -3a+2b \\ 0 & 1 & 2a-b \\ 0 & 0 & 7a-5b+c \end{array} \right) \quad \Rightarrow 7a-5b+c=0$$

Q) Show that the vectors $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ span $\mathbb{R}^2$ over $\mathbb{R}$

Let $a, b \in \mathbb{R}$

Let $v = a v_1 + b v_2$ C.L.C. of vectors in question

$$v = \begin{pmatrix} a \\ a \end{pmatrix} + \begin{pmatrix} b \\ 2b \end{pmatrix} = \begin{pmatrix} a+b \\ a+2b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix} \text{ (say)}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix} \equiv \left(\begin{array}{cc|c} 1 & 1 & p \\ 1 & 2 & q \end{array} \right) \quad R_2' = R_2 - R_1$$

$$\equiv \left(\begin{array}{cc|c} 1 & 1 & p \\ 0 & 1 & q-p \end{array} \right) \quad R_1' = R_1 - R_2$$

$$\equiv \left(\begin{array}{cc|c} 1 & 0 & 2p-q \\ 0 & 1 & q-p \end{array} \right) \quad \begin{array}{l} a = 2p-q \\ b = q-p \end{array}$$

$\exists a, b \in \mathbb{R} \quad \forall p, q \in \mathbb{R} \Rightarrow \vec{v}_1, \vec{v}_2 \text{ span } \mathbb{R}^2 \text{ over } \mathbb{R}$
 (ie any $\vec{v} \in \mathbb{R}^2$)
 $(2p-q \in \mathbb{R} \text{ and } q-p \in \mathbb{R} \because p, q \in \mathbb{R})$

* LINEAR INDEPENDENCE, BASIS, DIMENSION

Q) $f(x) = 2x^3 + x^2 + x + 1$
 $g(x) = x^3 + 3x^2 + x - 2$
 $h(x) = x^3 + 2x^2 - x + 3$

$$a f(x) + b g(x) + c h(x) = 0$$

$$\Rightarrow x^3 (2a + b + c) + x^2 (a + 3b + 2c) + x (a + b - c) + 1(a - 2b + 3c) = 0$$

Comparing coeff we get

$$\text{Aug} = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -2 & 3 & 0 \end{array} \right)$$

$$R_2' = R_2 - R_1/2 \quad R_3' = R_3 - \frac{R_1}{2} \quad R_4' = R_4 - \frac{R_1}{2}$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 0 & 5/2 & 3/2 & 0 \\ 0 & 1/2 & -3/2 & 0 \\ 0 & -5/2 & 5/2 & 0 \end{array} \right)$$

$$\begin{aligned} R_2' &= 2R_2 \\ R_3' &= 2R_3 \\ R_4' &= -2/5 R_4 \\ R_1' &= R_1/2 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 1/2 & 1/2 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right)$$

$$\begin{aligned} R_4' &= R_4 - R_3 \\ R_2' &= R_2 + R_3 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 1/2 & 1/2 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right)$$

$$\begin{aligned} \Rightarrow c &= 0 \\ b &= 0 \\ a &= 0 \end{aligned}$$

$a = b = c = 0 \Rightarrow$ Linear Independence

Q) Check Linear Independence of $f(x), g(x)$

$$f(x) = x \quad g(x) = e^{2x}$$

$$\text{Let } af(x) + bg(x) = 0 \\ ax + be^{2x} = 0$$

$$f = x \\ g = e^{2x}$$

Wronskian Determinant (if $= 0$, l.d else l.i)

$$W = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} x & e^{2x} \\ 1 & 2e^{2x} \end{vmatrix} = (2x-1)e^{2x} \neq 0 \text{ in gen}$$

Hence, f and g are linearly independent

Now, $h(x) = \sin x$

$$|W| = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix} = \begin{vmatrix} x & e^{2x} & \sin x \\ 1 & 2e^{2x} & \cos x \\ 0 & 4e^{2x} & -\sin x \end{vmatrix}$$

$$= x[-2\sin x e^{2x} - 4e^{2x}\cos x] - [-\sin x e^{2x} - 4e^{2x}\sin x]$$

$$= e^{2x} [5\sin x - 2x(\sin x + 2\cos x)]$$

$\neq 0$ in general

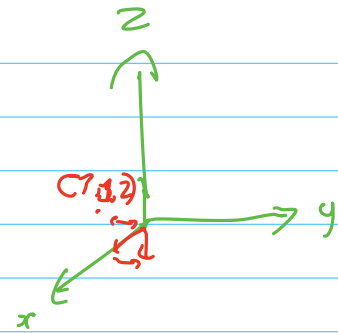
$\therefore$ LI

Q) Determine whether $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$

$v_3 = \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$, $v_4 = \begin{bmatrix} 2 \\ 6 \\ 8 \\ 5 \end{bmatrix}$ forms a basis of $\mathbb{R}^4$ over $\mathbb{R}$

If not, determine the dimension of the subspace they span.

$$C(3,1,2) = 3C(1,0,0) + 1C(0,1,0) + 2C(0,0,1)$$



Similarly
 $C(x_i, y_i, z_i)$

$$= x_i C(1,0,0) + y_i C(0,1,0) + z_i C(0,0,1)$$

↳ Real, so field is $\mathbb{R}$

Any random vector $C(x_i, y_i, z_i)$ can be written as a suitable L.C. of $C(1,0,0)$, $C(0,1,0)$, $C(0,0,1)$

↳ d.i. also

∴ $C(1,0,0)$, $C(0,1,0)$, $C(0,0,1)$ → basis

(d.i. and span the whole space)

— Basis is a linearly independent set of vectors which span the whole vector space.

Cardinality of basis = Dimension of the vector space

30/08/23

* NULL SPACE & RANGE SPACE

$$\begin{aligned} Q > \quad x + y &= 2 \\ x + 2y + z &= 4 \\ y + 2z + t &= 4 \\ z + 2t &= 3 \end{aligned}$$

$$\downarrow$$

$$(1, 1, 1, 1)$$

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 2x_1 + 3x_2 + x_3 &= 2 \\ 3x_1 + 6x_2 + 4x_3 &= 1 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 2 \\ 3 & 6 & 4 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 1 \\ 1 & 2 & 0 & -3 \end{array} \right]$$

↳ No soln

Augmenting it with these vectors

$$\left[\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 1 & 2 & 1 & 0 & 0 & 4 & b_2 \\ 0 & 1 & 2 & 1 & 0 & 4 & b_3 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 0 & 1 & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 1 & 2 & 1 & 0 & 4 & b_3 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c|c|c} 1 & 1 & 0 & 0 & 0 & 2 & b_1 \\ 0 & 1 & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 0 & 1 & 1 & 0 & 2 & b_3 - b_2 + b_1 \\ 0 & 0 & 1 & 2 & 0 & 3 & b_4 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cccc|c|c|c} 1 & \underline{1} & 0 & 0 & 0 & \underline{2} & \underline{b_1} \\ 0 & 1 & 1 & 0 & 0 & 2 & b_2 - b_1 \\ 0 & 0 & 1 & 1 & 0 & 2 & b_3 - b_2 + b_1 \\ 0 & 0 & 0 & \textcircled{1} & 0 & 1 & b_4 - b_3 + b_2 - b_1 \end{array} \right]$$

rank = 4

→ Homogeneous

If we wanted to solve $\underline{A}\underline{x} = \underline{0}$, we'll do a simple back substitution

$$t = 0$$

$$z + t = 0 \Rightarrow z = 0$$

$$y + z = 0 \Rightarrow y = 0$$

$$x + y = 0 \Rightarrow x = 0$$

Here, $(0, 0, 0, 0)$ is the only solⁿ for $\underline{A}\underline{x} = \underline{0}$. Hence $\underline{A}\underline{x} = \underline{b}$ will have a unique solⁿ.

$$\text{For } \underline{A}\underline{x} = \underline{b}$$

$$t = 1$$

$$z + t = 2 \Rightarrow z = 1$$

$$y + z = 2 \Rightarrow y = 1$$

$$x + y = 2 \Rightarrow x = 1$$

unique solⁿ

Now, for $\underline{A}\underline{x} = \underline{b}$, does the solⁿ exist $\forall \underline{b}$? Will we always get a solution?

For Q2

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & 3 & 1 & 0 \\ 5 & 6 & 4 & 0 \end{array} \right] \left[\begin{array}{c|c} 1 & b_1 \\ 2 & b_2 \\ 1 & b_3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right] \left[\begin{array}{c|c} 1 & b_1 \\ 0 & b_2 - 2b_1 \\ -4 & b_3 - 5b_1 \end{array} \right]$$

$$\text{rank} = 2 \leftarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -4 \end{array} \right] \left[\begin{array}{c|c} 1 & b_1 \\ 0 & b_2 - 2b_1 \\ 0 & b_3 - b_2 - 3b_1 \end{array} \right]$$

Now, for $\underline{Ax} = \underline{0}$

$$x_2 - x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

$$\# \text{ eqns} = 2$$

$$\# \text{ var} = 3$$

$$\text{DOF} = 1$$

$$\text{Let } x_2 = \alpha$$

$$\Rightarrow x_3 = \alpha$$

$$x_1 = -2\alpha$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

You have non zero, infinitely many solutions for $\underline{Ax} = \underline{0}$. You can generate infinite solutions by scalar multiplication of α .

$\Rightarrow$ All solutions belong to the linear vector space of $(2, 1, 1)$ ^{basis}
 $\downarrow$
 dimension = 1.

This space is called "Null" Space.

When $\text{rank}(A) < n$, we solve

- Condition for solvability: $b_3 - b_2 - 3b_1 = 0$.

$$\text{rank}(A) = 2$$

$\text{rank}(A|b)$ should be 2.

$$\# \text{ eqns} = 1$$

$$\# \text{ variable} = 3$$

$$\text{DOF} = 2$$

$$\Rightarrow \text{Let } b_1 = \beta, b_2 = \gamma \Rightarrow b_3 = 3\beta + \gamma$$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} \beta \\ \gamma \\ 3\beta + \gamma \end{bmatrix} = \beta \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

dimension = 2

This vector space, for vector b , which renders your system of equations solvable $\Rightarrow$ Range space

Solutions will never be unique.

Q) $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 2 \\ 3 & -1 & 9 \\ 2 & 1 & 1 \end{bmatrix}$ $b = \begin{bmatrix} 3 & 0 & 0 & 2 \end{bmatrix}^T$

4 eq's, 3 variables

$$\left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 2 & 1 & 2 & 0 & 0 & b_2 \\ 3 & -1 & 9 & 0 & 0 & b_3 \\ 2 & 1 & 1 & 0 & 2 & b_4 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 1 & 2 & 0 & 0 & -3 & b_2 - b_1 \\ -5 & 3.5 & 0 & 0 & -13.5 & b_3 - 4.5b_1 \\ 1.5 & 1.5 & 0 & 0 & 0.5 & b_4 - 0.5b_1 \end{array} \right]$$

$$\downarrow \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 1 & 2 & 0 & 0 & -3 & b_2 - b_1 \\ -6.75 & 0 & 0 & 0 & -8.25 & b_3 - 4.5b_1 - 1.75b_2 \\ 0.75 & 0 & 0 & 0 & 2.75 & b_4 - 0.5b_1 - 0.75b_2 \end{array} \right]$$

Clearly for $\underline{Ax=0} \Rightarrow x = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T$

$$\begin{aligned} b_3 - 4.5b_1 - 1.75b_2 &= 0 \\ b_4 - 0.5b_1 - 0.75b_2 &= 0 \end{aligned}$$

$$b_3 + b_4 - 5b_1 - 3.5b_2 = 0$$

$$\Rightarrow 3b_4 + b_3 - 4b_2 = 2b_1 = 0$$

$$\text{rank} = 4$$

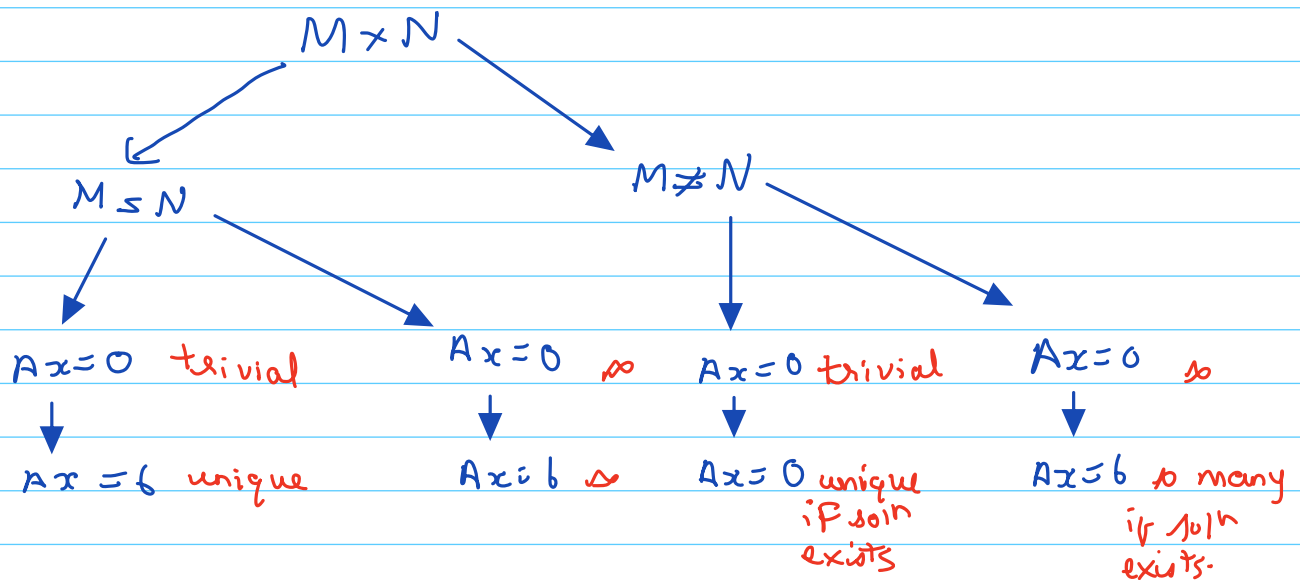
$$\epsilon_{\epsilon}^n = 1 \Rightarrow \text{POF} = 3$$

$$\text{Let } b_1 = \alpha, b_2 = \beta, b_4 = \gamma \Rightarrow b_3 = 2\alpha + 2\beta - 3\gamma$$

$$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

Range space. $\text{Pdim} = 3$

If b can be written as a L.C. of the range space
 $\Rightarrow$ solvable.



$$\dim(\text{range space}) + \dim(\text{null space}) = \dim(\text{space we are dealing with})$$

Note: zero vector has no dimension

6/9/23

LINEAR TRANSFORMATION

$$V_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad V_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \quad \alpha \in F$$

$$V_3 = \alpha \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad V_4 = \alpha \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + b \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \times \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

- Linear transformation maps two new different Linear vector spaces.

$$t: \mathbb{R}^{2 \times 1} \rightarrow \mathbb{R}^{3 \times 1}$$

- A set of polynomial & differential operator $\rightarrow$ operator (Algebraic)
- $S = \{ax: a \in \mathbb{R}, a > 0\}$ & differential operator

Transformation
 $\rightarrow$
 $S = \{a: a \in \mathbb{R}, a > 0\}$

Linearity $\rightarrow \underline{v_1}, \underline{v_2}, \alpha$

$$\begin{aligned} \textcircled{1} \quad t(\underline{v_1} + \underline{v_2}) &= t(\underline{v_1}) + t(\underline{v_2}) \\ \textcircled{2} \quad t(\alpha \underline{v_1}) &= \alpha t(\underline{v_1}) \end{aligned} \quad \left. \begin{aligned} & \\ & \end{aligned} \right\} \begin{aligned} &t(\alpha \underline{v_1} + \beta \underline{v_2}) \\ &= \alpha t(\underline{v_1}) + \beta t(\underline{v_2}) \end{aligned}$$

Q> Let V be a set of continuous differential function with the following operator defined:

$$T: V \rightarrow V$$

$$T[f(x)] = \int_0^x f(t) dt \quad \forall f(x) \in V; x \in \mathbb{R}$$

Identify if its Linear or not?

Let $f(x)$ & $g(x) \in V$

$$T[f(x)] = \int_0^x f(t) dt \quad T[g(x)] = \int_0^x g(t) dt$$

$$\begin{aligned} \textcircled{1} \quad T[f(x) + g(x)] &= \int_0^x \{f(t) + g(t)\} dt \\ &= \int_0^x f(t) dt + \int_0^x g(t) dt \\ &= T[f(x)] + T[g(x)] \end{aligned}$$

$$\textcircled{2} \quad T[\alpha f(x)] = \int_0^x \alpha f(t) dt = \alpha \int_0^x f(t) dt = \alpha T[f(x)]$$

Linear

Q> Identify if they are Linear or not

$$t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$(a) \quad t(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z)$$

$$(b) \quad t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$t(x, y, z) = (x + 1, y + 1)$$

$$\begin{aligned} (a) \quad \text{Let } v_1 = (x_1, y_1, z_1) \in \mathbb{R}^3 &\Rightarrow \alpha v_1 = (\alpha x_1, \alpha y_1, \alpha z_1) \\ v_2 = (x_2, y_2, z_2) &\Rightarrow \alpha v_2 = (\alpha x_2, \alpha y_2, \alpha z_2) \end{aligned}$$

$$\Rightarrow t(v_1) = (x_1 + 2y_1 - 3z_1, 4x_1 - 5y_1 + 6z_1)$$

$$t(v_2) = (x_2 + 2y_2 - 3z_2, 4x_2 - 5y_2 + 6z_2)$$

$$\alpha v_1 + \beta v_2 = (\underbrace{\alpha x_1 + \beta x_2}_{x_3}, \underbrace{\alpha y_1 + \beta y_2}_{y_3}, \underbrace{\alpha z_1 + \beta z_2}_{z_3})$$

$$\Rightarrow t(\alpha v_1 + \beta v_2) = [\alpha x_1 + \beta x_2 + 2(\alpha y_1 + \beta y_2) - 3(\alpha z_1 + \beta z_2), 4(\alpha x_1 + \beta x_2) - 5(\alpha y_1 + \beta y_2) + 6(\alpha z_1 + \beta z_2)]$$

$$= (x_3 + 2y_3 - 3z_3, 4x_3 - 5y_3 + 6z_3)$$

$$t(\alpha\theta_1) = (\alpha x_1 + 2\alpha y_1 - 3\alpha z_1, 4\alpha x_1 - 5\alpha y_1 + 6\alpha z_1)$$

$$t(\beta\theta_2) = (\beta x_2 + 2\beta y_2 - 3\beta z_2, 4\beta x_2 - 5\beta y_2 + 6\beta z_2)$$

$$\forall x_1, x_2 \in \mathbb{R} \quad \therefore \alpha x_1 + \beta x_2 \in \mathbb{R}$$

Based on above $\rightarrow$ Linear

$$\langle b \rangle \quad \text{Let } \theta_1 = (x_1, y_1, z_1) \in \mathbb{R}^3 \Rightarrow \alpha\theta_1 = (\alpha x_1, \alpha y_1, \alpha z_1) \\ \theta_2 = (x_2, y_2, z_2) \quad \beta\theta_2 = (\beta x_2, \beta y_2, \beta z_2)$$

$$\alpha\theta_1 + \beta\theta_2 = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2)$$

$$t(\alpha\theta_1 + \beta\theta_2) = (\alpha x_1 + \beta x_2 + 1, \alpha y_1 + \beta y_2 + 1) \\ = (x_3 + 1, y_3 + 1)$$

$$t(\alpha\theta_1) = (\alpha x_1 + 1, \alpha y_1 + 1)$$

$$t(\beta\theta_2) = (\beta x_2 + 1, \beta y_2 + 1)$$

$\Rightarrow$ not linear

$$a) \quad \hat{L} = \frac{d^2}{dx^2} - m^2$$

$$b) \quad \hat{L} = \frac{\partial}{\partial t} - q \frac{\partial^2}{\partial z^2} + p \rightarrow \text{conservation.}$$

a) Let f_1 & f_2 be two continuous functions $\in V$

$$\hat{L} f_1 = \frac{d^2 f_1}{dx^2} - m^2 f_1 \quad \hat{L} f_2 = \frac{d^2 f_2}{dx^2} - m^2 f_2$$

$$\begin{aligned}
 \hat{L}(f_1 + f_2) &= \frac{d^2}{dx^2}(f_1 + f_2) - m^2(f_1 + f_2) \\
 &= \frac{d^2 f_1}{dx^2} - m^2 f_1 + \frac{d^2 f_2}{dx^2} - m^2 f_2 \\
 &= \hat{L} f_1 + \hat{L} f_2
 \end{aligned}$$

$$\begin{aligned}
 \hat{L}(\alpha f_1) &= \frac{d^2}{dx^2}(\alpha f_1) - m^2(\alpha f_1) \\
 &= \alpha \frac{d^2 f_1}{dx^2} - \alpha m^2 f_1 = \alpha \left(\frac{d^2 f_1}{dx^2} - m^2 f_1 \right) \\
 &= \alpha \hat{L} f_1
 \end{aligned}$$

$\Rightarrow$ Linear

b) Let $f_1(t, z)$ and $f_2(t, z)$ be two continuous functions EV

$$\hat{L} f_1 = \frac{\partial f_1}{\partial t} - \alpha \frac{\partial^2 f_1}{\partial z^2} + \beta$$

$$\hat{L} f_2 = \frac{\partial f_2}{\partial t} - \alpha \frac{\partial^2 f_2}{\partial z^2} + \beta$$

$$\hat{L}(f_1 + f_2) = \frac{\partial (f_1 + f_2)}{\partial t} - \alpha \frac{\partial^2 (f_1 + f_2)}{\partial z^2} + \beta$$

$$= \left(\frac{\partial f_1}{\partial t} - \alpha \frac{\partial^2 f_1}{\partial z^2} + \beta \right) + \left(\frac{\partial f_2}{\partial t} - \alpha \frac{\partial^2 f_2}{\partial z^2} + \beta \right)$$

$\hookrightarrow$ Not Linear $\curvearrowright$

c) Let $B = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\} \subset \mathbb{R}^3(\mathbb{R})$

$$t: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad t \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Determine $t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

$$\alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \xrightarrow{\text{ER}}$$

$$\Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 1 & -1 & 1 & z \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 0 & -1 & 2 & x+z \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} -1 & 0 & 1 & x \\ 0 & 1 & -1 & y \\ 0 & -1 & 1 & x+y+z \end{array} \right]$$

$$\Rightarrow x+y+z = \gamma$$

$$y = \beta - \gamma$$

$$-\alpha + \gamma = x$$

$$\Rightarrow y+z = \alpha$$

$$x+2y+z = \beta$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \left\{ \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow t \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} y+z \\ x+2y+z \\ x+y+z \end{bmatrix}$$

$$\Rightarrow t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$